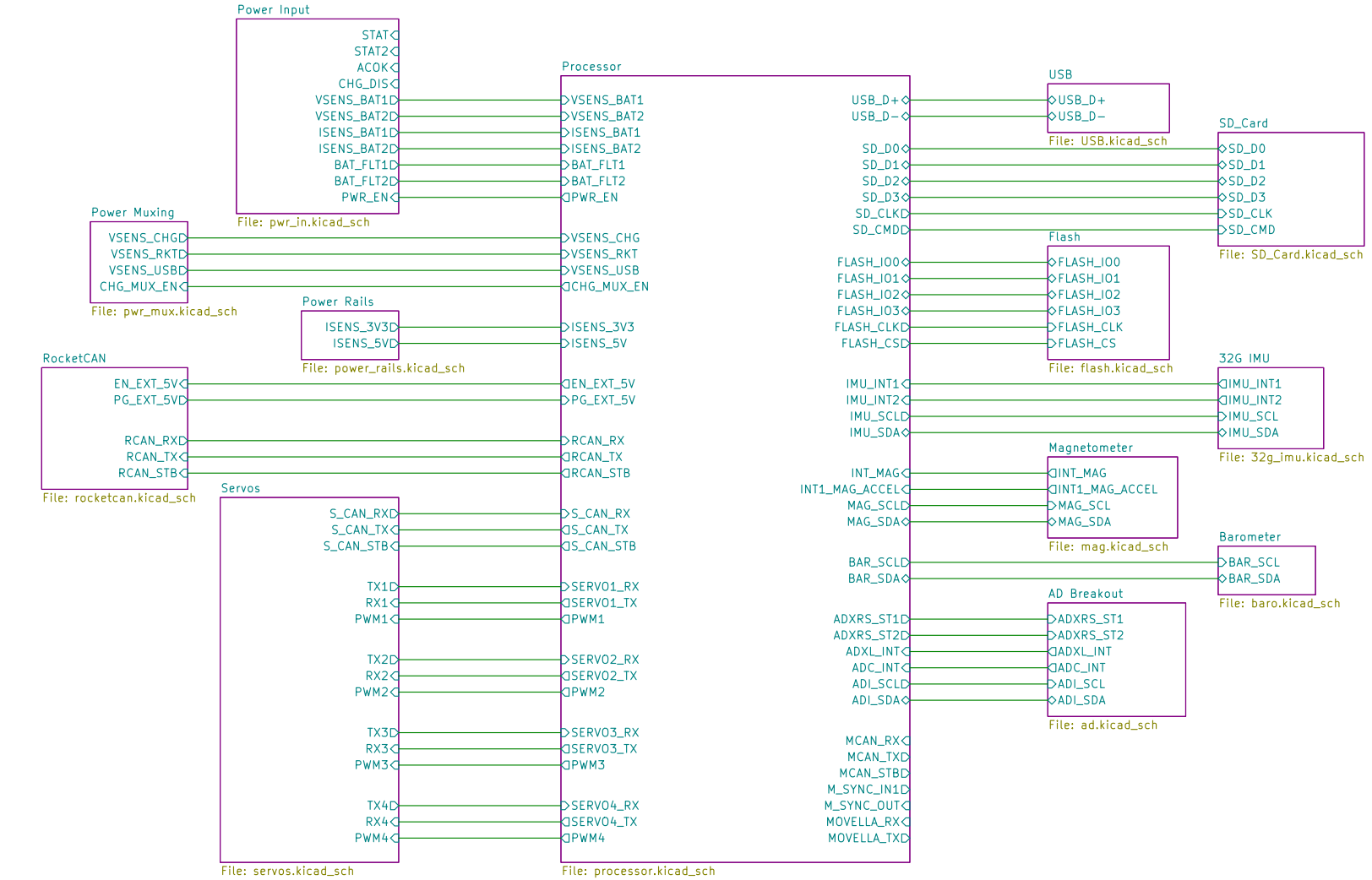


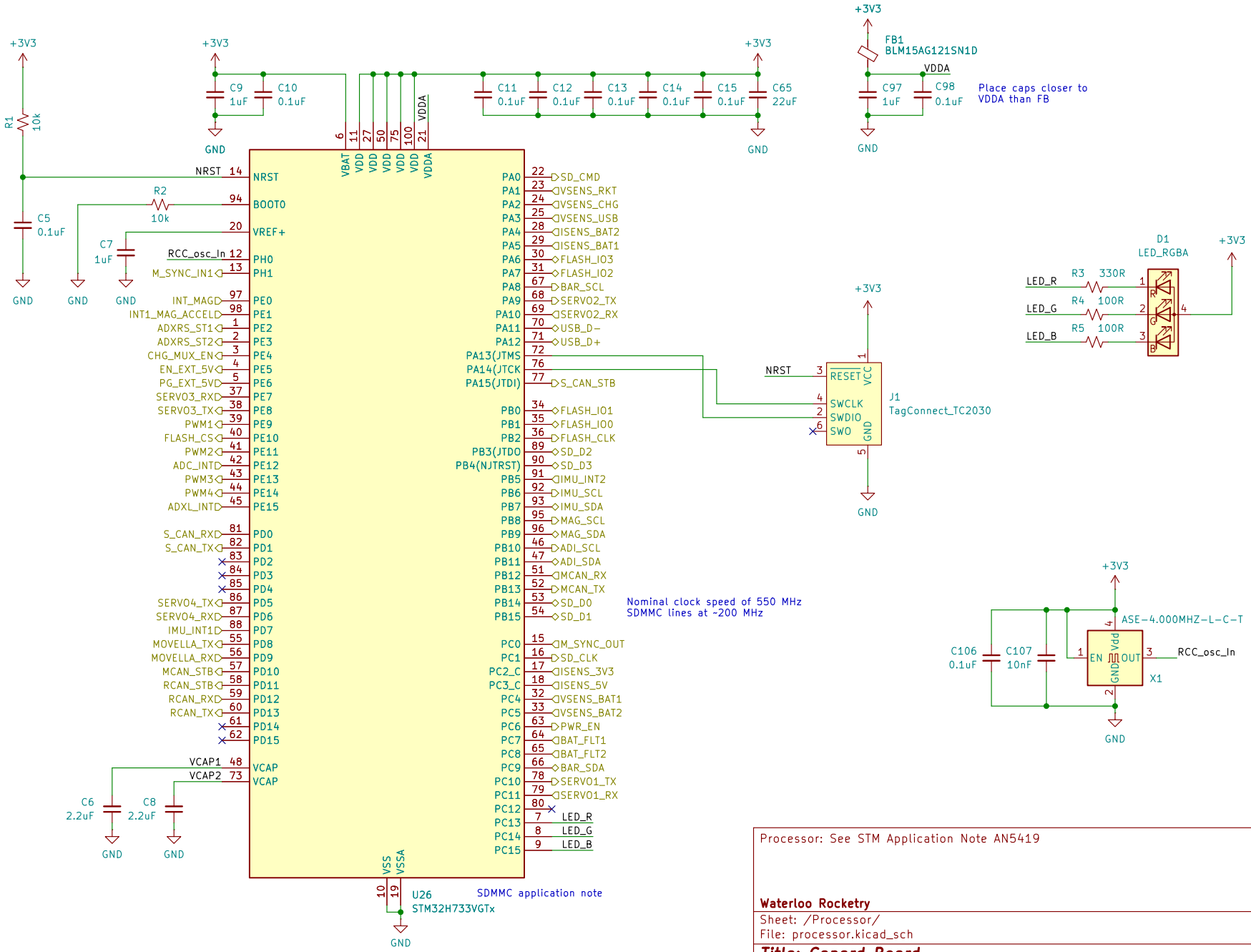
Processor: See STM Application Note AN5419

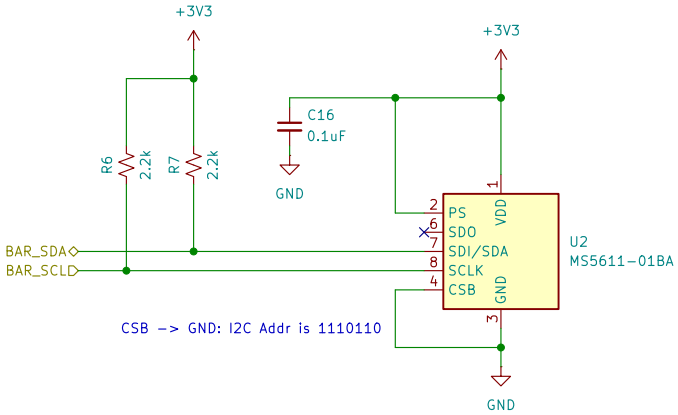
Waterloo Rocketry

Sheet: /
File: CanardBoard.kicad_sch

Title: Canard Board

Size: A4	Date: 2026-03-19	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 1/14


+3V3
R1
10k
C9
1uF
C10
0.1uF
GND
VBAT
VDD
VDD
VDD
VDD
VDD
VDDA
6
11
27
50
75
100
21
C11
0.1uF
C12
0.1uF
C13
0.1uF
C14
0.1uF
C15
0.1uF
C65
22uF
FB1
BLM15AG121SN1D
VDDA
C97
1uF
C98
0.1uF
Place caps closer to
VDDA than FB
NRST 14
R2
10k
94
BOOT0
20
VREF+
C5
0.1uF
C7
1uF
RCC_osc_In 12
M_SYNC_IN1 13
INT_MAGD 97
INT1_MAG_ACCELD 98
ADXRS_ST1 1
ADXRS_ST2 2
CHG_MUX_EN 3
EN_EXT_5V 4
PG_EXT_5VD 5
SERVO3_RXD 37
SERVO3_TX 38
PWM1 39
FLASH_CS 40
PWM2 41
ADC_INTD 42
PWM3 43
PWM4 44
ADXL_INTD 45
S_CAN_RXD 81
S_CAN_TX 82
83
84
85
SERVO4_TX 86
SERVO4_RXD 87
IMU_INT1D 88
MOVELLA_TX 55
MOVELLA_RXD 56
MCAN_STB 57
RCAN_STB 58
RCAN_RXD 59
RCAN_TX 60
61
62
VCAP1 48
VCAP2 73
C6
2.2uF
C8
2.2uF
PE0
PE1
PE2
PE3
PE4
PE5
PE6
PE7
PE8
PE9
PE10
PE11
PE12
PE13
PE14
PE15
PD0
PD1
PD2
PD3
PD4
PD5
PD6
PD7
PD8
PD9
PD10
PD11
PD12
PD13
PD14
PD15
VCAP
VCAP
PA0 22
PA1 23
PA2 24
PA3 25
PA4 28
PA5 29
PA6 30
PA7 31
PA8 67
PA9 68
PA10 69
PA11 70
PA12 71
PA13(JTMS 72
PA14(JTCK 76
PA15(JTDI) 77
PB0 34
PB1 35
PB2 36
PB3(JTDO 89
PB4(NJTRST) 90
PB5 91
PB6 92
PB7 93
PB8 95
PB9 96
PB10 46
PB11 47
PB12 51
PB13 52
PB14 53
PB15 54
PC0 15
PC1 16
PC2_C 17
PC3_C 18
PC4 32
PC5 33
PC6 63
PC7 64
PC8 65
PC9 66
PC10 78
PC11 79
PC12 80
PC13 7
PC14 8
PC15 9
PA0 SD_CMD
PA1 VSENS_RKT
PA2 VSENS_CHG
PA3 VSENS_USB
PA4 ISENS_BAT2
PA5 ISENS_BAT1
PA6 FLASH_IO3
PA7 FLASH_IO2
PA8 BAR_SCL
PA9 SERVO2_TX
PA10 SERVO2_RX
PA11 USB_D−
PA12 USB_D+
PA13(JTMS SWDIO
PA14(JTCK SWCLK
PA15(JTDI) S_CAN_STB
PB0 FLASH_IO1
PB1 FLASH_IO0
PB2 FLASH_CLK
PB3(JTDO SD_D2
PB4(NJTRST) SD_D3
PB5 IMU_INT2
PB6 IMU_SCL
PB7 IMU_SDA
PB8 MAG_SCL
PB9 MAG_SDA
PB10 ADI_SCL
PB11 ADI_SDA
PB12 MCAN_RX
PB13 MCAN_TX
PB14 SD_D0
PB15 SD_D1
PC0 M_SYNC_OUT
PC1 SD_CLK
PC2_C ISENS_3V3
PC3_C ISENS_5V
PC4 VSENS_BAT1
PC5 VSENS_BAT2
PC6 PWR_EN
PC7 BAT_FLT1
PC8 BAT_FLT2
PC9 BAR_SDA
PC10 SERVO1_TX
PC11 SERVO1_RX
PC12
PC13 LED_R
PC14 LED_G
PC15 LED_B
Nominal clock speed of 550 MHz
SDMMC lines at ~200 MHz
VSS
VSSA
10
19
U26
STM32H733VGTx
SDMMC application note
NRST
RESET
VCC
SWCLK
SWDIO
SWO
GND
J1
TagConnect_TC2030
D1
LED_RGBA
LED_R
LED_G
LED_B
R3 330R
R4 100R
R5 100R
C106
0.1uF
C107
10nF
ASE−4.000MHZ−L−C−T
EN
Vdd
GND
OUT
RCC_osc_In
X1
Processor: See STM Application Note AN5419
Waterloo Rocketry
Sheet: /Processor/
File: processor.kicad_sch
Title: Canard Board
Size: A4
Date: 2026−03−19
Rev: A
KiCad E.D.A. 10.0.7
Id: 2/14



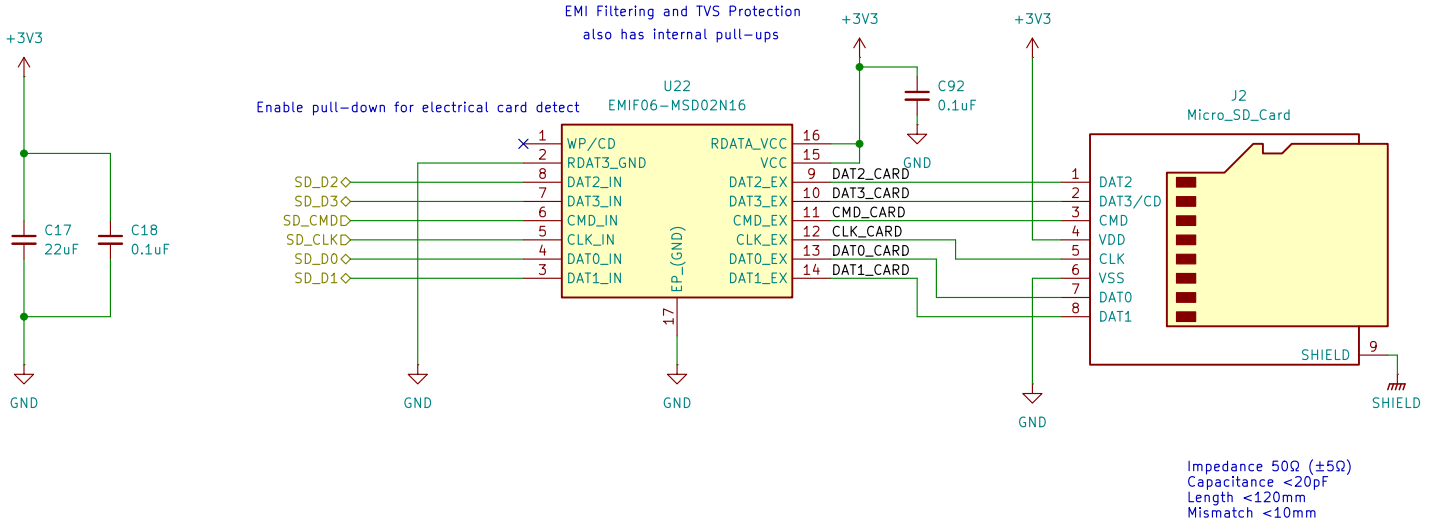
Processor: See STM Application Note AN5419

Waterloo Rocketry

Sheet: /Barometer/
File: baro.kicad_sch

Title: Canard Board

Size: A4	Date: 2026-03-19	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 3/14



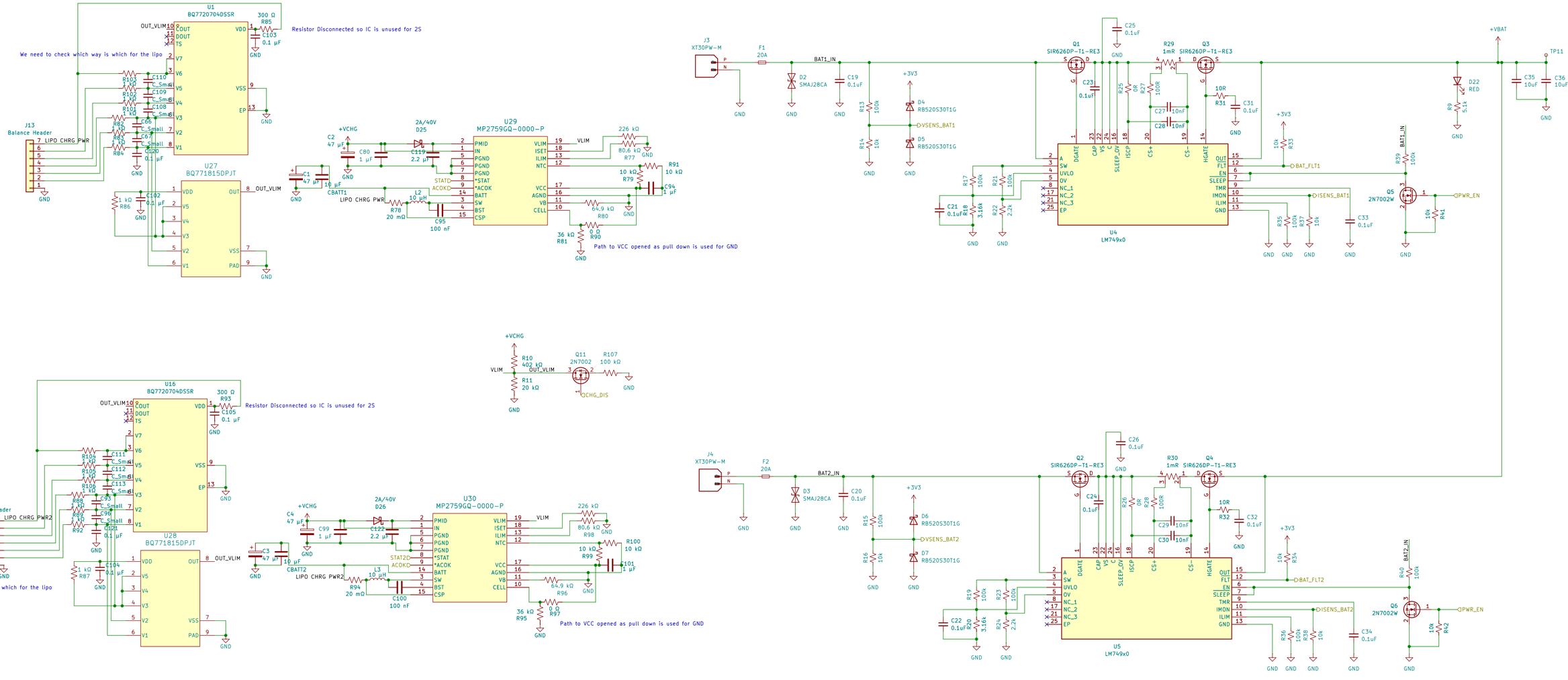
Processor: See STM Application Note AN5419

Waterloo Rocketry

Sheet: /SD_Card/
File: SD_Card.kicad_sch

Title: Canard Board

Size: A4	Date: 2026-03-19	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 4/14



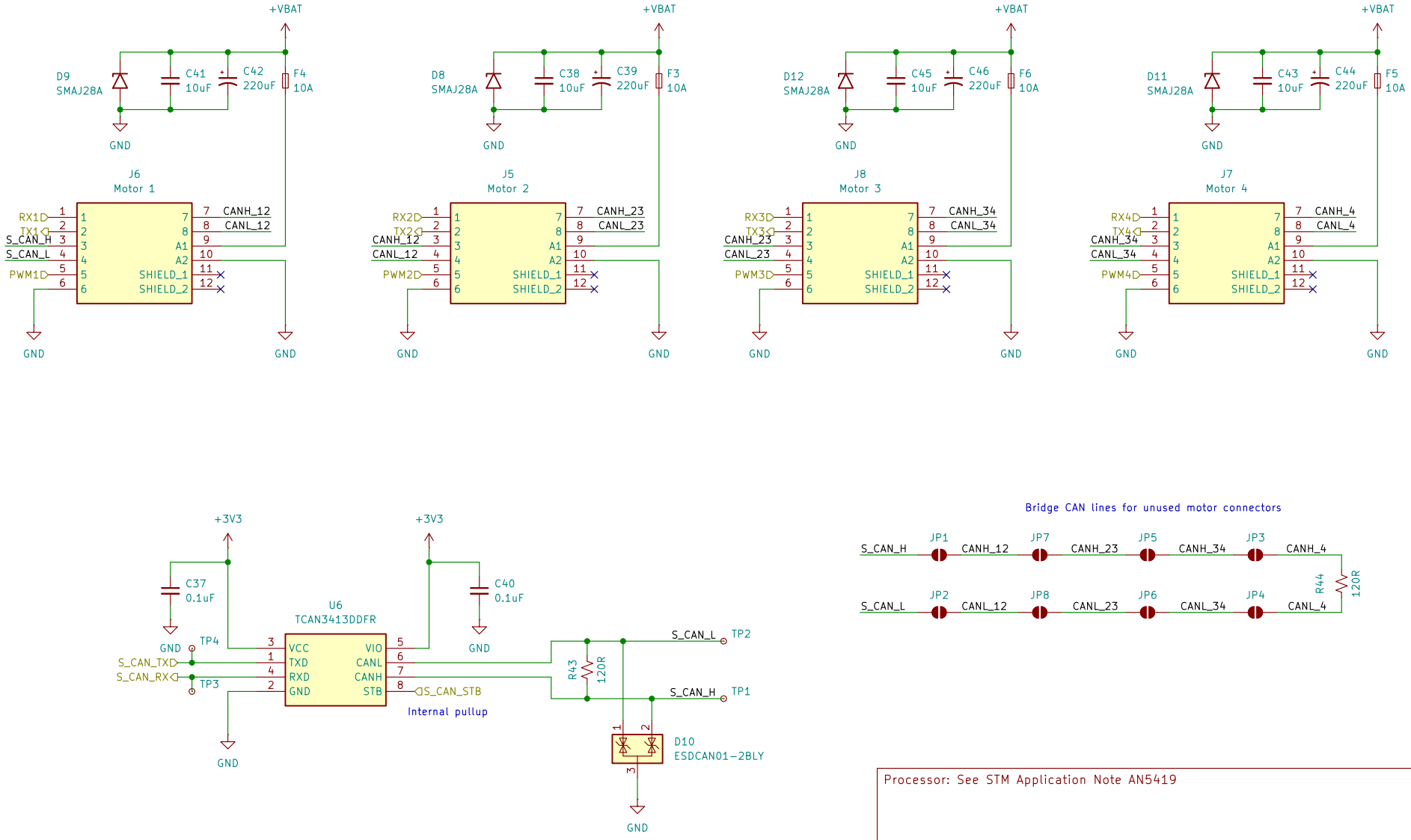
Processor: See STM Application Note AN5419

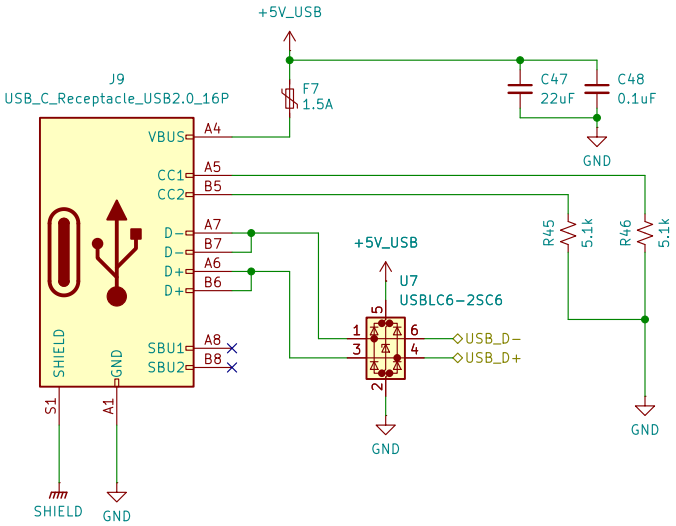
Waterloo Rocketry

Sheet: /Power Input/
File: pwr_in.kicad_sch

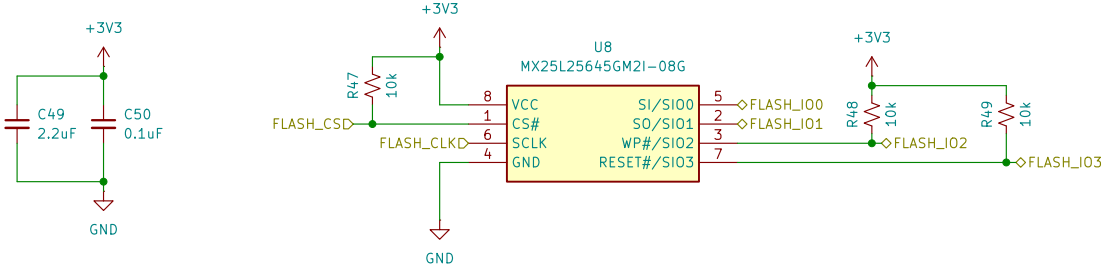
Title: Canard Board

Size: A2	Date: 2026-03-19	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 5/14

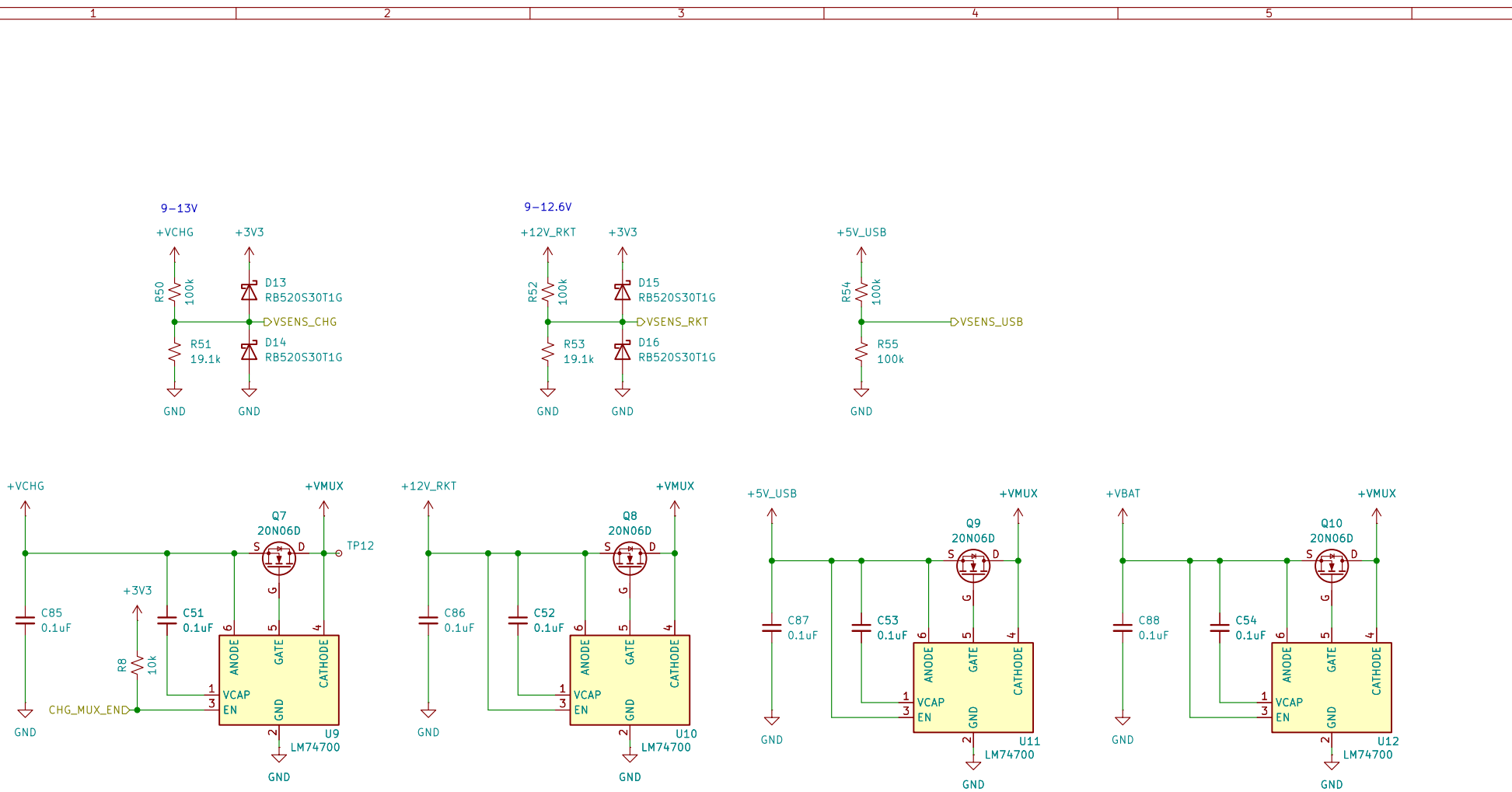

+VBAT
D9
SMAJ28A
C41
10uF
C42
220uF
F4
10A
GND
J6
Motor 1
RX1
TX1
S_CAN_H
S_CAN_L
PWM1
CANH_12
CANL_12
SHIELD_1
SHIELD_2
D8
C38
C39
F3
J5
Motor 2
RX2
TX2
CANH_12
CANL_12
PWM2
CANH_23
CANL_23
D12
C45
C46
F6
J8
Motor 3
RX3
TX3
CANH_23
CANL_23
PWM3
CANH_34
CANL_34
D11
C43
C44
F5
J7
Motor 4
RX4
TX4
CANH_34
CANL_34
PWM4
CANH_4
CANL_4
+3V3
C37
0.1uF
C40
U6
TCAN3413DDFR
VCC
TXD
RXD
GND
VIO
CANL
CANH
STB
S_CAN_TX
S_CAN_RX
S_CAN_STB
TP4
TP3
TP2
TP1
R43
120R
Internal pullup
D10
ESDCAN01-2BLY
Bridge CAN lines for unused motor connectors
JP1
JP2
JP7
JP8
JP5
JP6
JP3
JP4
R44
Processor: See STM Application Note AN5419
Waterloo Rocketry
Sheet: /Servos/
File: servos.kicad_sch
Title: Canard Board
Size: A4
Date: 2026-03-19
Rev: A
KiCad E.D.A. 10.0.7
Id: 6/14



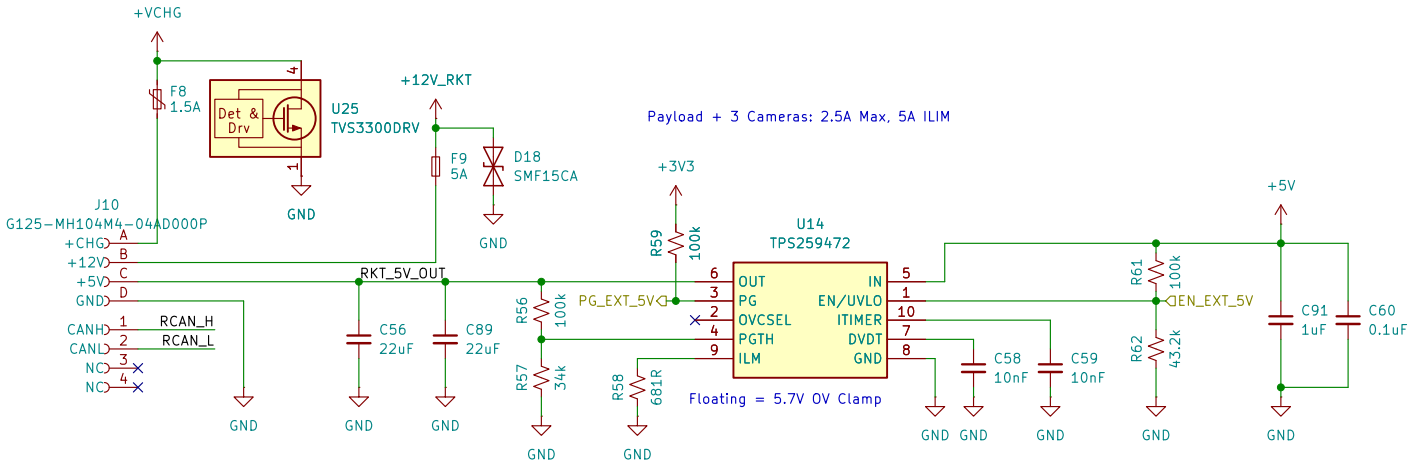
Sheet: /USB/		
File: USB.kicad_sch		
Title:		
Size: A4	Date: 2026-03-19	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 7/14

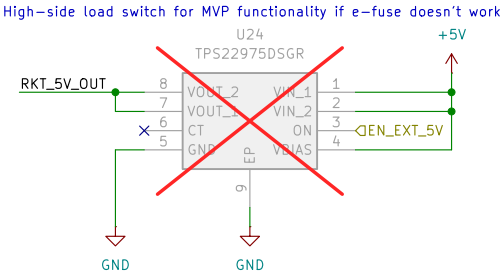


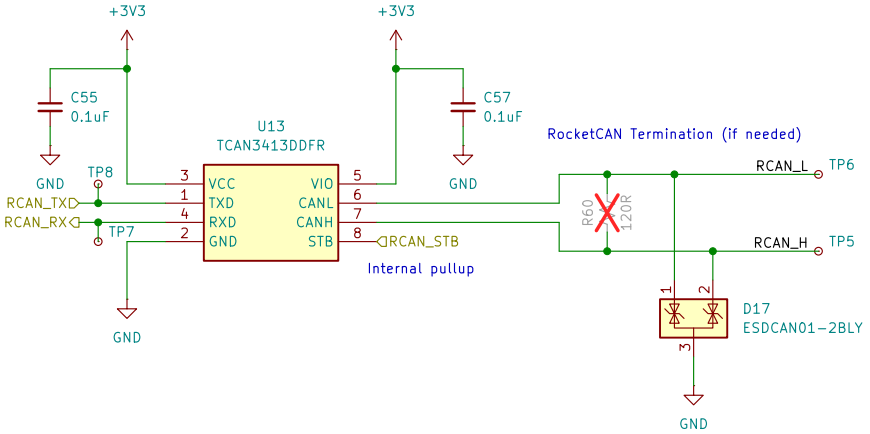
Sheet: /Flash/		
File: flash.kicad_sch		
Title:		
Size: A4	Date: 2026-03-19	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 8/14



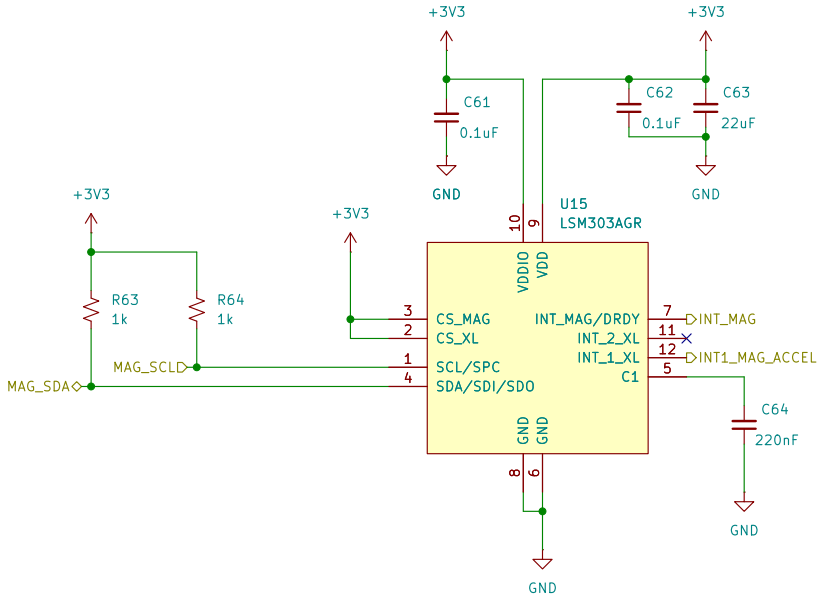
Sheet: /Power Muxing/		
File: pwr_mux.kicad_sch		
Title:		
Size: A4	Date: 2026-03-19	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 9/14





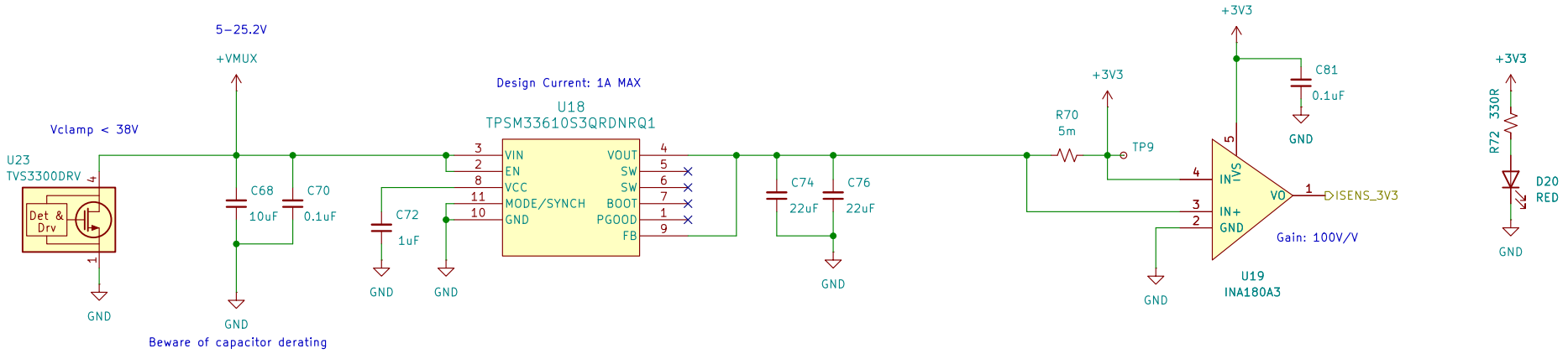


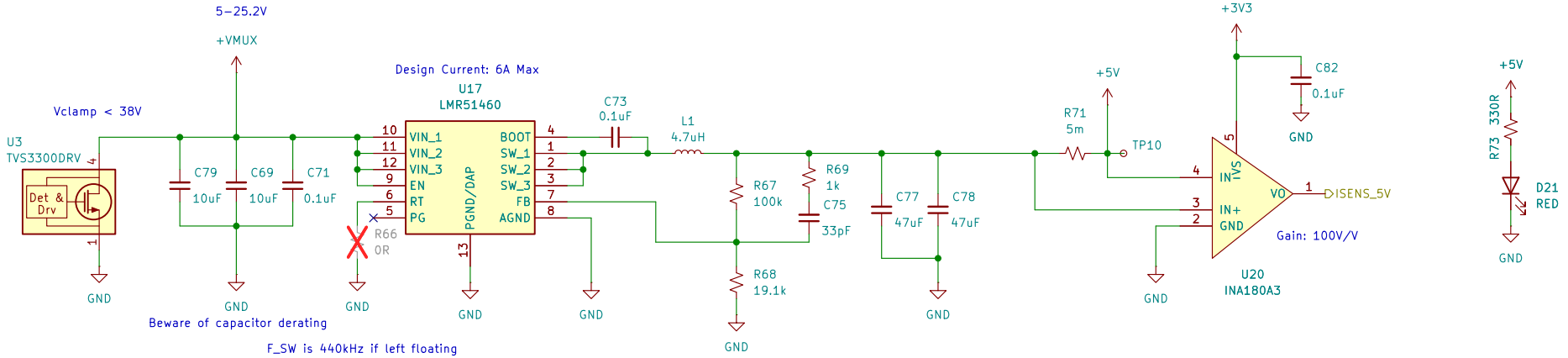
Sheet: /RocketCAN/		
File: rocketcan.kicad_sch		
Title:		
Size: A4	Date: 2026-03-19	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 10/14



IIS2MDC also works, just doesn't have the accel

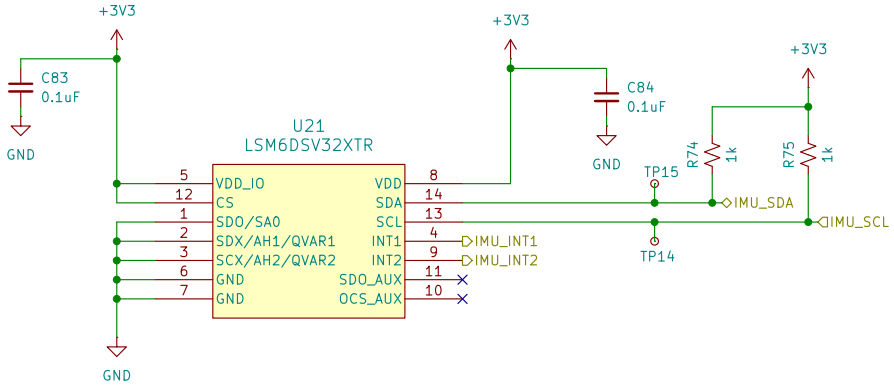
Sheet: /Magnetometer/ File: mag.kicad_sch		
Title:		
Size: A4	Date: 2026-03-19	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 11/14



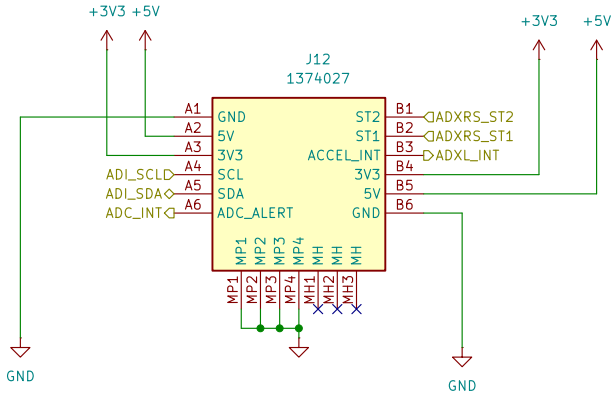


https://webench.ti.com/appinfo/webench/scripts/SDP.cgi?ID=BC8C0EE24EA233AC

Sheet: /Power Rails/		
File: power_rails.kicad_sch		
Title:		
Size: A4	Date: 2026-03-19	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 13/14



Sheet: /32G IMU/		
File: 32g_imu.kicad_sch		
Title:		
Size: A4	Date: 2026-03-19	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 14/14



I2C pullups on breakout board

Sheet: /AD Breakout/		
File: ad.kicad_sch		
Title:		
Size: A4	Date: 2026-03-19	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 15/14